

INFORMATION IN COMPLEX SYSTEMS

In inviting to participate in Venture Science project, we can read: „since over 50 years there was nothing revolutionary in science”. That's true. Although newest science has many interesting and very useful discoveries, there was in last 50 years no idea comparable with Relativity Theory or Quantum Physics.

There are many reasons of this state of things but as main and deepest of them we can show – exhaustion of traditional cognitive model. The main limitation of the model is the fact that not Physics nor Biology do not take into account – importance of Information in Nature. Saying it simply „Modern Physics deals with information carriers but not with Information itself”. In the meantime since the same 50 years mankind rapidly goes into Information Era. Newest inventions as generally available radio, TV, and internet have fundamentally change our life.

Research for Information are not shocking news: first works in this area was created long time ago. The most known manifest of new point of view was „The Turning Point” written by F. Capra in 1982. But in modern Physics, Biology and Medicine last traditional bio-mechanical model.

The conclusion is evident: the future of Science is to research for deep nature of Information. Maybe more: including Information in fundamental concepts of new cognitive model. The model does not exist now but is easy to imagine in near future. That's the reason for our suggestion that we named INFORMATION IN COMPLEX SYSTEMS.

As „complex system” we can consider many phenomena in Nature: from systems of elementary particles in atom, through systems of atoms in matter, live organisms interpreted as very complex informational systems up to ecosystems or even whole Universe. The last of presented visions, better known as *strong hypothesis of Gaia*, may be a main key to explain many phenomena actually called „paranormal”. If we want to maintain Science as most valuable knowledge of mankind, if we want to defend rationally point of view against commonplace actually irrationalism, we must include all the paranormal phenomena into Science. When we recognize Information as one of basic element of Universe (the remaining may be Energy and Timespace), there were nothing any „para”, all visible around us phenomena will stay „normal” and as normal they will be available for rational, critical analysis. This way Science can maintain its mission, i.e. make the knowledge accessible to every man who accept commonly rules of cognition.

The next benefit will be opening human perception beyond limitation of Quantum Physics. If we accept Copenhagen Interpretation of Quantum Physics instead of negate it, we get new giant intact area for researches for many years. The part of this New Science will be not „pure” physics, because integral part of it will be an observer and his psychic. So the New Great Unification we see rather as Unification of Nature and Aware Mind than actual imaginations of unification of four

known physical influences.

Recognition of Information and its importance in Universe may end the dangerous „schizma” which started in Science after appearances of three great „anarchists”: Thomas S. Kuhn, Paul K. Feyerabend and Carl R. Popper. Since the appearances there is in Science two fractions: the orthodox one tries to simulate that nothing happened, the modern one is thinking of implications of the appearances: what should Science do to adapt to permanent alternation. - The pride of Science is delivering knowledge, *best and surest at the time the knowledge was discovered*. Some scientists forgot these words and they want to make Science similar to a religion, propagating „eternal” truth. Recognizing the role of Information probably makes them aware that all things including Information are variable, so there is no sense to establish dogmas and then defend them.

The basic researches for Information will bear many fruits not only in theoretical knowledge; right now we may observe many tries to apply useful technologies based on information in medicine and psychology; probably from long time past hidden researches for information weapons. So all the more is including Information into mainstream of Science – for many profits for all of us.